The Relationship between Coercion and Protest

AN EMPIRICAL EVALUATION IN THREE COERCIVE STATES

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The relationship between coercion and protest, arguably the core of any theory of rebellion, remains unresolved. Competing hypotheses have emerged from formal models and empirical research. This article uses two forms of the predator-prey model to test these competing hypotheses. Time-series data from three coercive states (the German Democratic Republic, Czechoslovakia, and the Palestinian Intifada) are used to estimate parameters for both models. Results show stable, damped relationships in all three cases. The "inverted U" hypothesis receives less support than its "backlash" alternative, that is, that dissidents react strongly to extremely harsh coercion. Moreover, the study indicates that protesters adapt to coercion by changing tactics.

Recent revolutions have highlighted difficulties with the theories of protest, revolution, and political violence. Almost all of the underlying theorizing and empirical tests are cross-national and consequently cannot account for the interactive nature of coercion and protest. Some efforts have attempted to capture the dynamic nature of the process, but these typically have ignored each other and rarely confronted data (Lichbach 1992). Empirical testing of these models entails an independent data collection program.[1] This article

1. This is particularly true given Brockett's (1992) challenge to the reliability and validity of the Taylor and Jodice (1983) *World Handbook* protest and coercion data.

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seeks to fill the gap that has grown between theory and data in the study of the relationship between coercion and protest.

The article first presents competing hypotheses on the interaction of coercion and protest. It then estimates two dynamic models on three time series representing rebellion against repressive regimes. Along the way it is necessary to sort out several related issues: What is protest? What is coercion? How should both be measured? How are both affected by adaptation over time?

The article investigates three cases to assess the interaction between protest and coercion. All three present contexts of severe coercion with moderation at the conclusion of two. They represent three fundamentally different patterns of protest and coercion. The cases are (1) the former German Democratic Republic (GDR) from 1986 through 1989, (2) Czechoslovakia during the same period, and (3) the Palestinian Intifada from 1987 through 1989. Weekly aggregations of the data provide a reasonably fine-grained observation of the interaction between coercion and protest, which arise and subside abruptly. These cases were selected because they represent states with high levels of coercion and a dissatisfied populace. Leaders organized these regimes hierarchically in an effort to preclude concerted action by challengers. Coercion varied greatly in form among the cases but was high in all. Dissidents had no choice but to act in concert against the overwhelming force of the regime. Thus the assumption that both regime and challengers are single actors is more valid in these kinds of regimes than in most others. The article, then, attempts to shed light on protest and coercion in three cases in particular and in one type of regime in general.

COMPETING HYPOTHESES

The relationship most central to the formulation of a theory of rebellion is probably the interaction between the exercise of state authority and the challenges it confronts. Most early attempts at theory brought in clusters of variables that made the coercion-protest mix difficult to discern (e.g., Eckstein 1964; Gurr 1970; Tilly 1978). Among earlier researchers, Hibbs (1973) and Jackson et al. (1978) specifically addressed the coercion-protest nexus. The problem has attracted more attention recently, particularly in the work of Lichbach (1987), Tsebelis and Sprague (1989), Mason and Krane (1989), Muller and Weede (1990, 1994), Karmeshu and Mahajan (1990), Hoover and Kowaleski (1992), Khawaja (1993), Opp and Voss (1993), and Opp (1994).

Recent work has not resolved the problem. Lichbach (1987) wondered, "Why have scholars theorized and reported that all possible curves fit the impact of repression and dissent?" (p. 293). Lichbach helped to resolve the puzzle in terms of strategic choice, but empirical work has perpetuated the muddle. The dominant view holds that protest is most likely when coercion is moderate, least likely when it is absent or severe. Hibbs (1973, chap. 6) tested both this "inverted-U" formulation and the conventional hypothesis of an inverse relationship. He found empirical support for neither in the case of collective protest. Working with cross-sectional data and an indicator for coercion that comprised only "negative sanctions," for example, censorship, Hibbs (1973) concluded that protest is spontaneous and "does not appear to be deterred by the knowledge that elites in the past have resorted to repression" (pp. 91-92). DeNardo's (1985) formal, rational action analysis predicts a curvilinear, inverted-U relationship. Muller and Weede (1990) confirmed this curve in a large-sample, cross-national test with political deaths as the indicator for political violence. Opp (1994) used data from interviews in Leipzig to test and confirm the inverted-U hypothesis.

Anomalies in the inverted-U hypothesis have emerged recently. Hoover and Kowaleski (1992) argued that different situations produce different relationships between protest and coercion. Khawaja (1993) used event history analysis to study the reaction of West Bank collective action to various indexed measures of repression. He found confirmation of the inverted-U relationship for arrests, but also found that harsh forms of repression *increased* protest.

The hypothesis of an inverted-U relationship implies that any state progressing from low or high coercion to midrange coercion would confront a substantial rise in protest. It predicts no mass protest in the cases considered below. For the most part, large-scale protest was infrequent in both the GDR and Czechoslovakia until the fall of 1989. The Palestinians in the occupied territories were relatively quiescent until 1976, but even then protested at levels far below those reached suddenly at the outbreak of the Intifada in 1987. The inverted-U curve hypothesis is silent on these sudden surges of protest. There is, however, another perspective.

Khawaja's (1993) finding that harsh repression accelerated protest conforms to Hibbs's contention that protest erupts without regard for past repression. These results indicate that the inverted-U curve might yield another rise in protest at the high end of coercion. In other words, the relationship between protest and coercion may be nonlinear. Tsebelis and Sprague (1989) presented the dynamics of a nonlinear model under the hypothesis that the relationship between protest and coercion oscillates (pp. 551-52). Thus the coercive policies that might suppress protest at one time *incite* it at another. Mason and Krane (1989) argued that extremely harsh coercion might reduce protest temporarily, but would likely increase dissidence in the long run, especially if it is applied indiscriminately.

MODEL 1: TSEBELIS-SPRAGUE

Tsebelis and Sprague (1989) provided a straightforward means to assess the shape of the protest-coercion curve (pp. 553-54). They adapt the biological predator-prey model to the problem of revolution and coercion. The predator-prey model is the most relevant standard dynamic model for protest and coercion. Like the lynx and the rabbits of the original ecological model, the number of protesters (prey) depends, in part, on the level of coercion (predation) (see May 1973; Vandermeer 1981). The predator-prey model is better suited to study the interaction of coercion and protest than are existing protest-specific models. First, it is a dynamic model with a mechanism that closely replicates the pursuit of protesters by police. Second, unlike the most sophisticated modern protest formulations (e.g., Lichbach 1987), the model provides information on both the direction and the magnitude of change when protest and coercion rise and fall. Finally, the mathematics of the model have been deeply explored over a period of decades in a wide range of applications (see Canale 1970; May 1973; Boyce and DiPrima 1992). Tsebelis and Sprague present a canonical adaptation of the standard Lotka-Volterra model. Their most relevant model pares the predator-prey relationship to its elemental form:

$$\frac{dR}{dt} = -dR + gC + \beta \tag{1}$$

and

$$\frac{dC}{dt} = +hR - kC + y, \tag{2}$$

where

R = extent of revolutionary activity;
C = extent of state coercion; and
d, g, h, and k are parameters.

These linear equations have the scope to encompass the dynamic behavior that Tsebelis and Sprague explore. Coercion and protest might combine to produce a pattern of oscillation over time. The necessary and sufficient condition for such a pattern requires parameter estimates that generate a matrix with complex eigenvalues.

MODEL 2: LOTKA-VOLTERRA

The Lotka-Volterra predator-prey model offers an opportunity to estimate the coercion-protest relationship more directly through its inclusion of two interaction terms. The full predator-prey model generally is represented by the following equations:

$$\frac{dR(t)}{dt} = aR(t) - \alpha CR(t) \tag{3}$$

and

$$\frac{dC(t)}{dt} = -C(t)b + \beta CR(t), \tag{4}$$

where

a = rate at which protesters increase in the absence of coercion,
α = rate at which the interaction between protest and coercion decreases protest,
b = rate at which coercion declines in the absence of protest,
β = rate at which the interaction between protest and coercion increases coercion,
R = prey (protesters),
C = predators (coercion), and
t = time.

THE PROBLEM OF ADAPTATION

Protesters engage in an intermittent contest with the state. Lichbach (1987) showed that dissidents will change tactics over time, especially after defeats. Hoover and Kowaleski (1992) noted that protesters are less experienced politically than the regime, less bureaucratic, and thus more likely to experiment with new strategies (p. 151). As episodes of battle proceed, however, both sides have the time and capability to adapt to the other's tactics and resources. What kind of adaptation is available to dissidents under coercion? DeNardo (1985) focused on programmatic flexibility. Yet a highly autocratic, coercive state tolerates no rivals. Programmatic flexibility offers little prospect for success in such a regime. Other options are more promising. Suppose a dissident group stages a street demonstration, only to be teargassed, clubbed, and ultimately arrested. The group has little incentive to go back to the street after such an experience. Worse, its mobilization capability erodes as word of the repression flows through the society. Is the group limited to ideological compromise or surrender? Dissidents have other choices: If street demonstrations incur danger, mobilize quietly among workers and use the strike weapon. If the strike brings severe coercion, find a refuge that offers mobilization potential, for example, a church (mosque)-based group.[2]

Adaptation complicates the analysis of the interaction between protest and coercion. This article deals with the problem by using two different measures of protest and coercion: (1) demonstrations and their coercive countermeasures (including preemption) only and (2) other forms of protest, including strikes, petitions, and church-based rallies and their associated coercion. In most regimes, if protesters are proficient adapters, parameter estimates should be smaller and less robust for the samples representing alternative forms of protest and their attendant coercion. The cases in this article, however, represent extraordinarily coercive states that focused their security forces on public protest, for example, demonstrations and rallies. All three regimes were fully capable of responding to other challenges, but these incidents were likely to forge a more even match between state and dissidents and therefore a better fit for an interactive model like predator-prey.

I estimate these models, converted to difference equations. If the inverted-U hypothesis fits the cases, then the parameter estimates should reflect strong discouragement of protest following coercion and strong encouragement of coercion following protest. If any case is unstable and oscillatory, its parameter estimates should generate a matrix with complex eigenvalues. Finally, if the parameter estimates indicate that protest is strongly boosted by coercion, then they confirm the expectations of Lichbach (1987), Mason and Krane (1989), and Khawaja (1993) that especially harsh coercive measures give rise to significant dissident backlash.

Hypotheses for coercive states fashioned from this previous research take the following forms that are testable with the two dynamic models:

1. *Inverted U*: Coercion depresses protest. Protest accelerates coercion.
2. *Nonlinear*: The matrix generated by the parameter estimates has complex eigenvalues.
3. *Backlash*: Extremely harsh coercion accelerates protest.
4. *Adaptation*: Parameter estimates for demonstrations are less robust than for nondemonstration protest in highly coercive states.

These hypotheses should be evaluated with basic characteristics of the three cases in mind. For example, predator-prey models should fit continuous

2. This pattern is evident not only in the three cases investigated here but in many others as well, for example, in the African National Congress (ANC)-initiated guerrilla attacks in South Africa on regime institutions after the massacre of scores of women at Sharpeville in March 1960.

protest (Intifada) better than episodic conflict (GDR and Czechoslovakia). Coercion in Eastern Europe was often preemptive. It was more reactive in the Intifada. These considerations permit more specific predictions for the outcomes of estimating the two dynamic models. The inverted-U hypothesis predicts no protest, thus no significant fit for any case. It is improbable that the nonlinear hypothesis would fit the GDR and Czechoslovakian cases, whereas the Intifada is a good candidate for oscillation. Similarly, backlash is most likely in the case of the Intifada, where Khawaja (1993) detected it when there were harsh arrests. The adaptation hypothesis is a more problematic platform for prediction. On one hand, dissidents adapt to elude coercion. On the other hand, in a coercive state, such adaptation may provide real interaction (instead of overwhelming repression) between protesters and state.

	Hypotheses							
	Inverted U		*Nonlinear*		*Backlash*		*Adaptation*	
Case	*Demon-stration*	*Other Protest*	*Demon-stration*	*Other Protest*	*Demon-stration*	*Other Protest*	*Demon-stration*	*Other Protest*
Czechoslovakia								X
GDR								X
Intifada			X		X		X	

CONCEPTS: PROTEST AND COERCION

Two concepts require operational definition for this research. *Protest* involves observed activities directed against the regime. Demonstrations, strikes, church-based rallies, and petitions are the principal instruments of open protest under coercive regimes. Most empirical studies of protest use deaths from state violence as an indirect indicator of *political violence* (e.g., Muller and Seligson 1987; Muller and Weede 1990).[3] For this research, protest and coercion are two quite distinct concepts. *Coercion*, then, involves the state's repressive reaction to public protest (or its attempt to prevent protest). The coercion most visible to the outside world includes the killing and wounding of dissidents as well as their arrest and incarceration. Highly coercive states extend repression far beyond these bounds in their effort to eliminate political challenges. Just as there is no means to observe private

3. It is unclear how political violence relates to rebellion. For example, in the German Democratic Republic (GDR), a cumulative total of one third of the population marched against the regime, with little violence and no deaths. Almost 70% of the Czechoslovak people acted openly against the regime, with only one death.

acts of protest, no single indicator of coercion can capture the full measure of its application to a population.[4]

DATA AND METHOD

Two problems confront the coder of data on protest and coercion in repressive regimes. First, regimes take pains to conceal protest and coercion from the outside world. Second, Western media begin to report only extraordinary events after many years of repression in any country. Such reporting fatigue necessitates the use of multiple, especially regional, sources (see Brockett 1992). The data for this article were coded from both regional and international sources.[5]

PROTEST

What is protest in an autocratic regime? Almost any act of defiance challenges the regime, including demonstrations, strikes, petitions, and membership in proscribed groups or church-based reform movements. Perhaps the best evidence of what constitutes protest is what the regime actively discouraged or repressed. This is a system-specific problem. Unauthorized demonstrations were proscribed in the GDR and Czechoslovakia, as were strikes. Demonstrations were technically legal in the occupied territories of the West Bank and the Gaza Strip until the outbreak of the Intifada brought martial law.

These coding dilemmas were resolved in conjunction with the adaptation problem (above). The protest variable is presented in two forms. The first

4. In Eastern Europe, the state regularly took children from dissident families, demoted highly educated dissidents to manual labor, and deprived the children of even passive dissidents from higher education. Czechoslovak authorities threw poison pellets into the gardens of signatories to Charter 77. In the occupied territories of the West Bank and Gaza Strip, coercion included detention without trial, burning the houses of suspected dissidents, exile, and the withholding of employment.

5. The sources for the GDR and Czechoslovak data are the Foreign Broadcast Information Service or FBIS (period 1986-1992, Eastern Europe), *Frankfurter Allgemeine Zeitung* or *FAZ* (period 1986-1992), Reuters News Service from Nexis (period 1986-1992), the *New York Times* or *NYT* (period 1987-1992), *Keesings Archive*, the *Chicago Tribune*, Bundesminister für innerdeutsche Beziehungen (1988-1989), Mitter and Wolle (1990), and Ramet (1991). *Deutschland Archiv* was an additional source for the GDR. Intifada data were coded from Palestine Human Rights Information Center (1987-1989), Reuters News Service from Nexis (period 1986-1992), FBIS (period 1986-1992, Middle East), *FAZ* (period 1986-1992), *NYT* (period 1987-1992), *Süddeutsche Zeitung* (period 1986-1989), *Jerusalem Post*, Gerner (1991), Nassar and Heacock (1990), Benvenisiti (1986), Benvenisiti and Khayat (1988), Roy (1986), and Central Bureau of Statistics (1992). Coding rules for the project are available from the author. The data and a codebook will be available as a Class V data set at the Interuniversity Consortium for Political and Social Research in Ann Arbor.

measures the number of people who appear on the streets in a protest demonstration. The second combines other forms of protest, principally strikes, petitions, and church-based activities (in Eastern Europe). Both of these variables are used in the estimations.

COERCION

A hallmark of authoritarian states is a well-financed and effective coercive capacity. This is true in all three cases. Although Israel is a democracy, it functioned as a coercive autocracy in the occupied territories. Only observable coercion is codable for regimes of this sort. For the most part, this is coercion induced by protest (or by the threat of protest). It does not include more directed sanctions, such as the removal of children, the destruction of housing, and the withdrawal of employment. The principal measures of coercion for Eastern Europe are arrests, injuries, and deaths. They are aggregated to form two variables. Only coercive acts against demonstrators are coded for the first variable, whereas all coercion against other forms of protest (including strikes, petitions, and church-group actions) constitutes the second. The data for coercion in the case of the Intifada are deaths of protesters. In terms of fatalities, the Intifada is by far the most coercive case in the analysis.

Difference equations derived from the two forms of the predator-prey models were used to estimate parameters and to determine the fit of the three time series. Three methods generated estimates for each time series. First, nonlinear ordinary least squares (OLS) treat the two equations discretely. Second, two-stage least squares uses instrumental variables to calculate parameter estimates and provides their standard errors directly. Finally, three-stage least squares goes further to check for cross-correlations between the two equations in the system (see Theil 1978; Goldberger 1991).

The predator-prey model reaches equilibrium under stable, long-term conditions. In two of the three cases for this article, exogenous system shocks are evident at the end of the time series. Therefore, tests are done only on the East European cases prior to the most active phase of their revolutions: for the GDR, from 1986 until September 4, 1989; for Czechoslovakia, from 1986 until November 6, 1989.

RESULTS

The parameter estimates for all equations are presented in the tables below. The first set presents the estimates for the Tsebelis-Sprague model

Tsebelis-Sprague Model:

$$\Delta R = dR_t + gC_t$$
$$\Delta C = hR_t - kC_t$$

TABLE 1
German Democratic Republic: 1986 to September 25, 1989

Parameter	*Demonstrations*	*Other Protest*
d	–0.6045*	–0.5748*
	(13.48)	(8.77)
g	0.1863	–0.3379
	(0.82)	(0.23)
h	0.0096	0.00039
	(0.65)	(0.12)
k	0.813*	0.849*
	(10.74)	(11.86)
Eigenvalues	$\lambda_1 = 0.81426059$	$\lambda_1 = -0.57475223$
	$\lambda_2 = -0.60576059$	$\lambda_2 = 0.84885123$

NOTE: $N = 194$; t statistics are in parentheses below each parameter estimate.
$*p < .05$.

$$\Delta R = dR_t + gC_t$$
$$\Delta C = hR_t - kC_t$$

TABLE 2
Czechoslovakia: 1986 to November 6, 1989 (three-stage least squares)

Parameter	*Demonstrations*	*Other Protest*
d	–0.9376*	0.1746*
	(8.71)	(2.0)
g	–0.559	–1.1972*
	(0.23)	(9.85)
h	–0.00002	0.1943*
	(0.00)	(2.27)
k	0.9994*	1.2196*
	(9.28)	(10.2)
Eigenvalues	$\lambda_1 = -0.937605772$	$\lambda_1 = 0.496130676$
	$\lambda_2 = 0.999405772$	$\lambda_2 = 0.8980483244$

NOTE: $N = 200$; t statistics are in parentheses below each parameter estimate.
$*p < .05$.

$$\Delta R = dR_t + gC_t$$
$$\Delta C = hR_t - kC_t$$

TABLE 3
Intifada (three-stage least squares)

Parameter	*Demonstrations*	*Strikes*
d	–0.8848*	–0.5794*
	(10.47)	(6.79)
g	348.5024*	49699*
	(3.9)	(3.55)
h	0.0002444*	0.000001848*
	(4.26)	(3.98)
k	0.40724*	–0.499*
	(6.71)	(6.54)
Eigenvalues	$\lambda_1 = 0.4701035$	$\lambda_1 = -0.23359976$
	$\lambda_2 = 0.9476635$	$\lambda_2 = -0.84499624$

NOTE: $N = 156$; *t* statistics are in parentheses below each parameter estimate.
*$p < .05$.

Lotka-Volterra Model:
$$\Delta R = aR_t - \alpha CR_t$$
$$\Delta C = bC_t + \beta CR_t$$

TABLE 4
German Democratic Republic:
1986 to September 25, 1989 (three-stage least squares)

Parameter	*Demonstrations*	*Other Protest*
a	–0.619*	–0.5752*
	(14.77)	(8.76)
b	–0.656	–0.4374
	(0.28)	(0.14)
α	–0.000092	0.002257
	(0.9)	(0.24)
β	–.0000629	–0.00269
	(.06)	(0.13)
Eigenvalues	$\lambda_1 = -0.6190975$	$\lambda_1 = -0.5734849$
	$\lambda_2 = 0.000034594$	$\lambda_2 = -0.0044193848$

NOTE: $N = 156$; *t* statistics are in parentheses below each parameter estimate.
*$p < .05$.

$$\Delta R = aR_t - \alpha CR_t$$
$$\Delta C = bC_t + \beta CR_t$$

TABLE 5
Czechoslovakia: 1986 to November 6, 1989

Parameter	*Demonstrations*	*Other Protest*
a	−0.904*	0.15281
	(4.8)	(1.78)
b	−0.9995*	0.17195*
	(4.81)	(2.04)
α	0.000115	0.00001*
	(0.31)	(9.8)
β	−.00000045	−0.00001*
	(0.03)	(10.13)
Eigenvalues	$\lambda_1 = -0.0001271$	$\lambda_1 = 0.15259291$
	$\lambda_2 = 0.9038741$	$\lambda_2 = -0.0000226792$

NOTE: $N = 200$; t statistics are in parentheses below each parameter estimate.
$*p < .05$.

$$\Delta R = aR_t - \alpha CR_t$$
$$\Delta C = bC_t + \beta CR_t$$

TABLE 6
Intifada: 1987-1989

Parameter	*Demonstrations*	*Strikes*
a	−1.1185*	−0.471*
	(8.89)	(3.21)
b	−0.538*	0.0000048*
	(4.46)	(6.6)
α	−0.0626*	0.0139
	(3.31)	(0.8)
β	0.0000374*	−0.000000673*
	(2.19)	(7.79)
Eigenvalues	$\lambda_1 = -1.14784$	$\lambda_1 = -0.00000053$
	$\lambda_2 = 0.0293775$	$\lambda_2 = -0.470596143$

NOTE: $N = 156$; t statistics are in parentheses below each parameter estimate.
$*p < .05$.

(equations 1 and 2) on the GDR, Czechoslovakia, and Intifada time series (see Tables 1-3). Tables 4, 5, and 6 display results for the Lotka-Volterra predator-prey model (equations 4 and 5). R^2 values are not reported for the three-stage least squares estimates because they are not meaningful (see Goldberger 1991, 372).

PARAMETER ESTIMATES

The Tsebelis-Sprague model generated identical values across all methods. One would expect the same numerical results across two- and three-stage least squares because the model is exactly identified (see Schmidt 1976, 212). Nonetheless, the OLS estimates match these. The more complex Lotka-Volterra model generated parameter estimates that vary across the three methods. In these cases, the three-stage least squares estimate is assumed to be the most accurate. These are the only parameter estimates shown in the tables.

The most important and consistent finding is that in all forms of the three cases the parameter estimates indicate stable systems. Not one satisfies the nonlinear hypothesis.[6] Damping is evident in all of the time series. Thus there is general stability in three cases of severe coercion. The fact that all cases reach equilibrium implies ongoing conflict in the absence of an exogenous system shock that could break the prevailing system values. Precisely such shocks came in the GDR and Czechoslovakia in late 1989.

DEMONSTRATIONS

Parameter estimates in the Tsebelis-Sprague model for demonstrations in the GDR and Czechoslovakia indicate small and insignificant interactive effects. Previous coercion tended to boost protest in the GDR, but to thwart it in Czechoslovakia. Yet the effects are slight, because there was relatively little protest until 1989. These were coercive systems that maintained quick, effective responsiveness or preemption until they were overwhelmed in 1989. The Intifada presents a different picture. Coercion (deaths of Palestinians) accelerates protest. The estimated values of g and h (equations 1 and 2) are more robust for the Intifada than for the other cases. Higher levels of coercion boosted protest and higher levels of protest increased coercion in the Intifada. This is a recipe for oscillation, but the parameter estimates indicate damping and stability.

6. The eigenvalues are real numbers for every equation. For any system to oscillate, its eigenvalues (latent roots that are solutions to the characteristic equation) must be complex (see Goldberg 1986, 151-52, 172).

The Lotka-Volterra model provided overall results consistent with the Tsebelis-Sprague estimates. The parameter estimates for the interactive terms for demonstrations and coercion in Eastern Europe are not statistically significant. The Intifada shows small increases both in protest and coercion as a result of their interaction (parameters and α and β). However, estimates for the Lokta-Volterra model also indicate falling levels of protest and coercion from time t to $t + 1$ (parameters a and b). All of the parameter estimates for the Intifada are statistically significant. The Intifada was a struggle with high levels of sustained conflict that provides a closer fit to the predator-prey model than does the episodic conflict in Eastern Europe.

ADAPTATION

The results indicate slightly better overall fit for the demonstration data than for other forms of protest in the GDR and the Intifada, but markedly better fit on adaptive protest in Czechoslovakia. What accounts for these differences? First, the GDR and Czechoslovakia organized their security forces principally to deter public protest. Demonstrations were either preempted or immediately halted through arrests, injuries, or both. There was much less alternative protest in the GDR, except for underground church meetings. A more active, adaptive protest movement grew in Czechoslovakia as a result of the 1968 Warsaw Pact invasion and the (1977) Charter 77 petition movement. These protests taxed Czechoslovak security organizations. It was a fair match, as reflected in the better fit for adaptive protest in the Czechoslovak case. The fit of the Intifada's strikes is nearly as close as that for its demonstrations. This is probably an artifact of the wider opportunity for demonstrations during strikes and a single indicator of coercion.

Protesters were able to discover alternative means of showing disapproval of the regime, usually without a proportional increase in coercion. The better fit of demonstration data in the cases of the GDR and the Intifada also might be a product of the difficulty of measuring coercion that does not develop in response to street demonstrations. Noteworthy, however, is the close fit of Czechoslovakia's alternative protest.

DISCUSSION

What do the results say about the relationship between protest and coercion? Because all of these cases were stable, two questions stand out: First, how do the inverted-U, backlash, and adaptation hypotheses fare? Second, is there any evidence that coercion deters future protest?

The standard inverted-U hypothesis is supported weakly in Czechoslovakia, marginally negated in the GDR, and invalid in the Intifada. The inverted-U hypothesis foresees no significant protest in these cases. Yet episodic and alternative protest erupted in Eastern Europe, and high levels of protest characterized the Intifada. The backlash hypothesis is supported weakly in the GDR and strongly confirmed in the Intifada. Palestinians continued to protest even under extremely harsh coercion. The fact that this (measured) coercion consisted wholly of killing protesters supports the notion that especially harsh coercion leads to new bursts of protest. One of the highest levels of Czechoslovak protest in 1989 followed the fatal shooting of a demonstrator. This is also what Khawaja (1993) found on the West Bank, and it may be what Hibbs (1973) described as "spontaneous" protest. The adaptive hypothesis achieved mixed results. Attempts to adapt protest were apparent in all three cases. GDR and Palestinian protesters suffered less coercion when they adapted, but Czechoslovak petition signers, strikers, and members of culture clubs met a vigilant foe in the state secret police. Still, the extraordinary fit of Czechoslovak adaptive protest and coercion indicates a lower level of repression than that suffered by demonstrators.

It is clear that protest and coercion are significantly interdependent in the Intifada, but the other two cases show little statistical significance in the interactive terms. Do we have two different mechanisms operating in these three cases? All are stable, all have high levels of coercion, but the protest-coercion link seems to have been disjoined in the GDR and Czechoslovakia. The nature of these regimes explains this finding to an extent. Jackson et al. (1978, 652) noted that Eastern Europe showed the capability to maintain effective coercion by continuing to obtain sophisticated equipment with a strong commitment to the maintenance of sufficient force. Thus coercive capability was continuously high in Eastern Europe until late 1989. It was able to thwart public protest by preemption, for example, arresting dissidents the day before a significant anniversary such as June 17 in the GDR or August 20 in Czechoslovakia.

Yet protest erupted at unprecedented levels in 1989. The exogenous shocks that brought on the protest are well-known. The Brezhnev doctrine was withdrawn in July, Hungary opened its border to Austria, and thousands of East Germans fled to the West. Once massive demonstrations erupted in the GDR (arguably the most conservative Warsaw Pact country), they followed in Czechoslovakia and the rest of the bloc.

What was the *internal* mechanism that brought this diachronic change? In other words, how did dissidents for the first time mobilize large numbers of citizens quickly, even when coercion persisted? They appear to have had something that DeNardo (1985) stressed: they had power in numbers. Tilly

(1978) has long argued that the more power a group has, the less it is likely to be repressed. For dissidents in a coercive state, power lies in numbers—large numbers of mobilized opponents of the regime. The GDR and Czechoslovakia could effectively repress protests below the level of 300,000 to 500,000 demonstrators in any large city. The threat of coercion was salient and credible. Only when the organizational strength and geographic distribution of dissidents exceeded the coercive capability of the regime did repression subside (see Francisco 1993).

The level of coercion in the former Soviet bloc approached totalitarian control, not in the sense of tremendous violence, but through the close monitoring of activities and the control of employment, housing, and education. It is a pattern that continues elsewhere in the world, for example, in Myanmar or North Korea. The Intifada is different. It arose under military occupation among Palestinians who were linguistically and culturally distinct from their Israeli occupiers. Israeli tactics evolved to higher levels of violence without evidence of effective control. Detention without trial sought to deprive the movement of its leaders. Yet new leaders emerged. The introduction of live ammunition and house burning brought international condemnation, created martyrs, and did not diminish the intensity of the protest.

Coercive regimes resist mass mobilization efforts. It is this rational resistance that lies at the heart of the inverted-U hypothesis. Lichbach (1987, 286) notes that rational regimes will repress a dissident group's *less* effective tactic, for example, petitions rather than strikes. The Czechoslovak government did attempt to repress adaptive tactics, especially petitions and culture groups. Nonetheless, over time, even rational regimes encounter periods of weakness. Dissidents can exploit these periods as opportunities for larger scale mobilization. Opp and Voss (1993) stressed this point of opportunism in their study of the GDR revolution. Over time, such opportunities arise, at least through relative changes of coercive capacity. Once successful, dissident movements develop power in numbers (DeNardo 1985). In Tilly's (1978) terms, mobilization transforms the conflict into a test of organizational strengths.

DEFINING THE CURVE OF REVOLUTION

"Which curve is the curve of revolution?" asked Tsebelis and Sprague (1989). Beyond the support for the inverted-U hypothesis, the answer has been elusive. Tsebelis and Sprague (1989) and Karmeshu and Mahajan (1990) developed dynamics that lead to a range of plausible curves. This article's test of the predator-prey model provides empirical clues about why no curve of revolution can be as simple as the inverted-U curve.

What curves are suggested in this investigation? First, backlash is evident in the Intifada and in the massive outbreak of protest after police killed a demonstrator in Prague in 1989. By its nature, backlash suggests a change in the slope of mobilization. A regime that exceeds contextual limits of coercion may provide a point of inflection where postevent mobilization increases sharply. As Tsebelis and Sprague (1989) noted, backlash behavior might drive the system out of equilibrium and into oscillation. To date, no one has found an actual protest that oscillates. Second, adaptation creates its own pattern. Feedback, a critical factor in adaptive models, is rapid in coercive states: was the activist neighbor fired, arrested, or injured? Adaptation is also facilitated by community (see Lichbach 1994). Remarkable in the three cases that constitute this study are the shared values of the people challenging extraordinarily harsh governments. When adaptive protest succeeds without attendant coercion, increased mobilization should result. Under these circumstances, one might find growth of protest bound only by the size of the mobilizable population. At such a point, the regime must adapt to demands or risk collapse. Waiting too long to adapt prolongs and accelerates protest. Few regimes can survive this level of intransigence (see Francisco 1993).

Are the curves that fit these mobilizations similar across cases? The phase-plane plots of all three were quite comparable—in each case a sharp retreat to zero with actual initial conditions drawn from equivalent dates. A more recent test in a democratic setting produced the same pattern (Francisco 1994). These results suggest that equilibrium is the outcome of the protest-coercion nexus in a broad range of domestic conflicts. The Intifada remained stable despite backlash and heavy resistance to harsh coercion. The East European cases progressed to a higher level of equilibrium through massive nonviolent mobilization. The traditional assumption (e.g., Karmeshu and Mahajan 1990) is that this discontinuous jump is from the level of hostility. In fact, higher levels of equilibrium should indicate lower levels of violence, at least where acceleration of mobilization is attainable. In general, protests may not grow as rapidly as theorists assume. Too seldom considered are inherent mobilization limitations, for example, geographic disbursement or significant dropouts over time—protest is a most disruptive activity both to the government and to dissidents' own lives.

Threshold models are sometimes advocated as an alternative means to investigate the curve of revolution (Granovetter 1978; Granovetter and Soong 1983). These models are almost never tested empirically because they demand data that are rarely available: how many people joined in actions after each individual acted (Granovetter and Soong 1983, 177). The predator-prey model makes a better analytic tool for defining the curve of revolutions. It is adaptable to fit conditions of tripartite conflict, such as Northern Ireland. Using the predator-prey model in empirical tests of this sort can move the field of domestic conflict research beyond the conundrum of model building in vacuo identified by Lichbach (1992): "The use of difference or differential equations to model protest and repression . . . has been the antithesis of scholarship: stagnation and not progress, ad hocery and not accumulation" (p. 341).

This article's exploration of three coercive states is a first step in the more systematic empirical investigation of the interaction between protest and coercion. The most fundamental puzzles remain in that nexus: How does protest arise under coercion? Is "spontaneous" protest induced principally by harsh coercion? If so, in what contexts is it most likely to occur? How is the level of "acceptable" coercion established, and why do regimes exceed it? These are matters that lie at the center of conflict studies (see Lichbach 1994, 32).

Data problems pose a significant obstacle to progress. Most existing, large-scale data sets are designed for cross-sectional research. However, this article and others like it show that these problems can be overcome, if not for all countries, at least for several cases that are theoretically relevant. In fact, data properly collected in the first instance can obviate the need to move to ever higher levels of statistical complexity. We can best move toward a general theory of protest and revolution if we start at the core of the dissidents' relationship with the state. Once we understand better what happens in this crucible, moving outward should be more productive than the past decades of academic research that focused most on exogenous factors across an array of countries.

REFERENCES

Benvenisiti, M. 1986. *The West Bank handbook*. Boulder, CO: Westview.

Benvenisiti, M., and S. Khayat. 1988. *The West Bank and Gaza atlas*. Jerusalem: The Jerusalem Post.

Boyce, W. E., and R. C. DiPrima. 1992. *Elementary differential equations*. New York: Wiley.

Brockett, C. D. 1992. Measuring political violence and land inequality in central America. *American Political Science Review* 86:169-76.

Bundesminister für innerdeutsche Beziehungen. 1988-89. *Informationen*. Bonn, Germany: Bundesminister für innerdeutsche Beziehungen.

Canale, R. 1970. An analysis of models describing predator-prey interaction. *Biotechnology and Bioengineering* 12:353-78.

Central Bureau of Statistics. 1992. *Statistical abstract of Israel*.

DeNardo, J. 1985. *Power in numbers*. Princeton, NJ: Princeton University Press.

Eckstein, H., ed. 1964. *Internal war: Problems and approaches*. New York: Free Press.

Francisco, R. 1993. Theories of protest and the revolutions of 1989. *American Journal of Political Science* 37:663-80.

———. 1994. Coercion and protest: An empirical test in two democratic states. Paper presented at the annual meeting of the American Political Science Association, September 1-4, New York.

Frankfurter Allgemeine Zeitung. FAZ 1986-1992.

Gerner, D. 1991. *One land, two peoples: The conflict over Palestine*. Boulder, CO: Westview.

Goldberg, S. 1986. *Introduction to difference equations*. New York: Dover.

Goldberger, A. 1991. *A course in econometrics*. Cambridge, MA: Harvard University Press.

Granovetter, M. 1978. Threshold models of collective behavior. *American Journal of Sociology* 83:1420-43.

Granovetter, M., and R. Soong. 1983. Threshold models of diffusion and collective behavior. *Journal of Mathematical Sociology* 9:165-79.

Gurr, T. R. 1970. *Why men rebel*. Princeton, NJ: Princeton University Press.

Hibbs, D., Jr. 1973. *Mass political violence: A cross-national causal analysis*. New York: Wiley.

Hoover, D., and D. Kowaleski. 1992. Dynamic models of dissent and repression. *Journal of Conflict Resolution* 36:150-82.

Jackson, S., B. Russett, D. Snidal, and D. Sylvan. 1978. Conflict and coercion in dependent states. *Journal of Conflict Resolution* 22:627-57.

Karmeshu, V. P. Jain, and A. K. Mahajan. 1990. A dynamic model of domestic political conflict process. *Journal of Conflict Resolution* 34:252-69.

Khawaja, M. 1993. Repression and popular collective action: Evidence from the West Bank. *Sociological Forum* 8:47-71.

Lichbach, M. 1987. Deterrence or escalation? The puzzle of aggregate studies of repression and dissent. *Journal of Conflict Resolution* 31:266-97.

———. 1992. Nobody cites nobody else: Mathematical models of domestic political conflict. *Defence Economics* 3:341-57.

———. 1994. Rethinking rationality and rebellion. *Rationality and Society* 6:8-39.

Mason, T. D., and D. Krane. 1989. The political economy of death squads: Toward a theory of the impact of state-sanctioned terror. *International Studies Quarterly* 33:175-98.

May, R. 1973. *Stability and complexity in model ecosystems*. Princeton, NJ: Princeton University Press.

Mitter, A., and S. Wolle, eds. 1990. *Ich liebe euch doch alle: Befehle und Lageberichte des MfS, Januar-November 1989* (But I love you all: Command and situation reports of the Ministry for State Security). Berlin: Basis Druck.

Muller, E., and M. Seligson. 1987. Inequality and insurgency. *American Political Science Review* 81:425-51.

Muller, E., and E. Weede. 1990. Cross-national variation in political violence. *Journal of Conflict Resolution* 34:624-51.

———. 1994. Theories of rebellion: Relative deprivation and power contention. *Rationality and Society* 6:40-57.

Nassar, J., and R. Heacock, eds. 1990. *Intifada: Palestine at the crossroads*. New York: Praeger.

Opp, K. 1994. Repression and revolutionary action: East Germany in 1989. *Rationality and Society* 6:101-38.

Opp, K., and P. Voss. 1993. *Die volkseigene Revolution*. Stuttgart, Germany: Klett-Cotta.

Ramet, S. 1991. *Social currents in Eastern Europe*. Durham, NC: Duke University Press.

Roy, S. 1986. *The Gaza Strip survey*. Boulder, CO: Westview.

Schmidt, P. 1976. *Econometrics*. New York: Marcel Dekker.

Taylor, C., and D. Jodice. 1983. *World handbook of political and social indicators*. New Haven, CT: Yale University Press.

Theil, H. 1978. *Introduction to econometrics*. Englewood Cliffs, NJ: Prentice-Hall.

Tilly, C. 1978. *From mobilization to revolution*. Reading, MA: Addison-Wesley.

Tsebelis, G., and J. Sprague. 1989. Coercion and revolution: Variations on a predator-prey model. *Mathematical and Computer Modelling* 12:547-59.

Vandermeer, J. 1981. *Elementary mathematical ecology*. New York: Wiley.